

Limits Drill Sheet

First Calculus Practice: Substitution, Algebra, Factoring, and One-Sided
Limits

300 progressively harder exercises: 50 per topic

Purpose

This drill packet gives first-time calculus students a visual introduction to what limits mean, then repeated practice recognizing the form of a limit, choosing the correct algebraic move, and deciding when a two-sided limit exists. Each topic has fifty problems in five levels.

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What a Limit Is

Main idea

A limit describes what output value a function approaches as the input slides toward a chosen number. We are not always asking for the value at the point. We are asking what value the function is getting close to near the point.

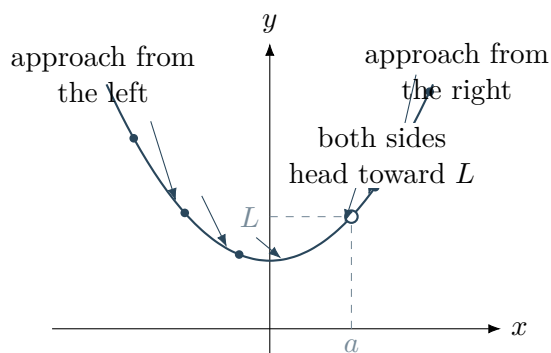
Imagine walking along the graph of a function. The symbol

$$\lim_{x \rightarrow a} f(x) = L$$

means: as your x -position moves closer and closer to a , your height on the graph moves closer and closer to L .

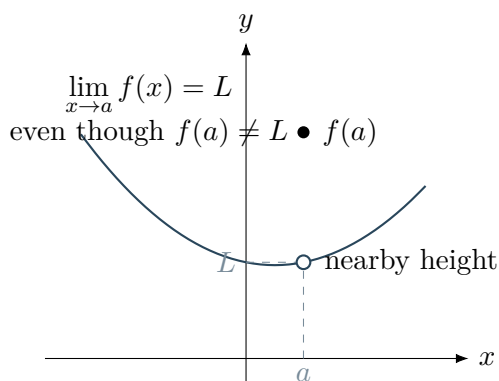
The function might actually be defined at $x = a$, or it might have a hole, a jump, or some other issue there. The limit cares about the behavior nearby. This is why a limit can exist even when $f(a)$ does not exist, and why a limit can fail to exist even when the function has values on both sides.

Sliding toward a point



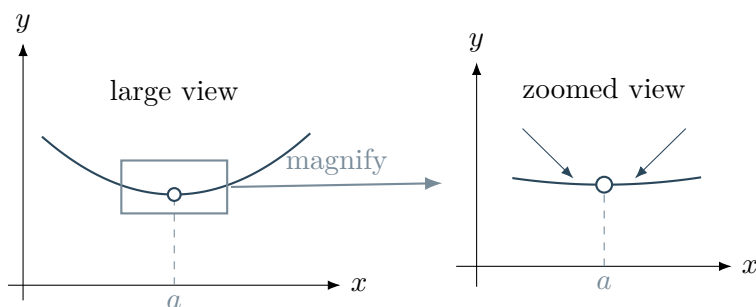
In this picture, the graph has a hole at the point above $x = a$. That hole does not automatically destroy the limit. If both sides of the graph slide toward the same height L , then the limit is L .

The value at the point is a separate question



This is one of the most important habits in limits: separate the value at the point from the value approached near the point. The filled dot shows $f(a)$. The open circle shows the height the curve is approaching. The limit follows the open-circle behavior, not the filled dot.

Zooming in



Zooming in is a good mental model. The closer x gets to a , the more the outputs cluster near one height. In later calculus, this idea supports infinitesimal thinking: over a tiny interval, a curve can often be approximated by something simpler. Limits are the language that makes that “tiny interval” idea precise.

When plug-and-chug is enough

If the expression is continuous at the target value, you may plug in the target value. This is not cheating. It is exactly what the limit laws allow. For example, polynomial expressions and rational expressions with nonzero denominators at the target value can be evaluated by substitution.

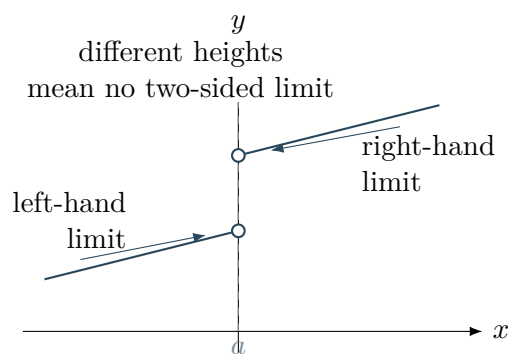
If substitution gives a normal number, stop there. The limit is that number.

When the sides matter

Sometimes the left side and right side behave differently. Then we use one-sided limits:

$$\lim_{x \rightarrow a^-} f(x) \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x).$$

The minus sign means approach from values less than a . The plus sign means approach from values greater than a .



Approaching from the sides is still perfectly legitimate limit work. If both one-sided limits agree, the two-sided limit exists. If they disagree, the two-sided limit does not exist.

Topic 1

Basic Plug-and-Chug Limits

Learning target

Evaluate limits of polynomial and simple algebraic expressions by direct substitution.

Teaching notes

A limit asks what value an expression approaches as x gets close to a chosen number. In this first type of problem, there is no hidden obstacle: the expression is built from powers, multiplication, addition, and subtraction, so it behaves smoothly near the target value.

Big idea

For polynomials and simple continuous expressions, the limit is found by replacing x with the number it approaches. The arrow $x \rightarrow a$ does not mean “ x equals a forever.” It means we are watching the expression as x gets closer and closer to a .

Procedure

1. Identify the number that x approaches.
2. Substitute that number into the expression.
3. Simplify carefully using the order of operations.

Common trap

Do not make this harder than it is. If substituting gives an ordinary real number, that number is the limit.

Worked example

Problem. $\lim_{x \rightarrow 2} (3x^2 - 4x + 1)$

Move. Substitute $x = 2$ because polynomials are continuous.

Result. $3(2)^2 - 4(2) + 1 = 5$.

Drill

Level I: direct recognition

1. $\lim_{x \rightarrow -3} x - 4$
2. $\lim_{x \rightarrow -2} 2x - 3$
3. $\lim_{x \rightarrow -1} 3x - 2$
4. $\lim_{x \rightarrow 0} 4x - 1$
5. $\lim_{x \rightarrow 1} x$
6. $\lim_{x \rightarrow 2} 2x + 1$
7. $\lim_{x \rightarrow 3} 3x + 2$
8. $\lim_{x \rightarrow 4} 4x + 3$
9. $\lim_{x \rightarrow -4} x + 4$
10. $\lim_{x \rightarrow 5} 2x + 5$

Level II: routine variations

1. $\lim_{x \rightarrow -3} x^2 + 2x - 3$
2. $\lim_{x \rightarrow -2} 2x^2 + x - 2$
3. $\lim_{x \rightarrow -1} 3x^2 - 1$
4. $\lim_{x \rightarrow 0} x^2 - x$
5. $\lim_{x \rightarrow 1} 2x^2 - 2x + 1$
6. $\lim_{x \rightarrow 2} 3x^2 + 2x + 2$
7. $\lim_{x \rightarrow 3} x^2 + x + 3$
8. $\lim_{x \rightarrow 4} 2x^2 - 3$
9. $\lim_{x \rightarrow -4} 3x^2 - x - 2$
10. $\lim_{x \rightarrow 5} x^2 - 2x - 1$

Level III: disguised or prepared forms

1. $\lim_{x \rightarrow -3} x^2 + 7x + 9$
2. $\lim_{x \rightarrow -2} x^2 + 6x + 3$
3. $\lim_{x \rightarrow -1} x^2 + 5x - 1$
4. $\lim_{x \rightarrow 0} x^2 + 4x - 3$
5. $\lim_{x \rightarrow 1} x^2 + 3x + 1$
6. $\lim_{x \rightarrow 2} x^2 - 3x + 3$
7. $\lim_{x \rightarrow 3} x^2 - 4x + 7$
8. $\lim_{x \rightarrow 4} x^2 - 5x + 13$
9. $\lim_{x \rightarrow -4} x^2 + 12x + 16$
10. $\lim_{x \rightarrow 5} x^2 - 5x + 24$

Level IV: multi-stage execution

1. $\lim_{x \rightarrow -3} x^3 + 2x - 7$
2. $\lim_{x \rightarrow -2} 2x^3 - x^2 + 3x - 7$
3. $\lim_{x \rightarrow -1} x^3 - 2x^2 + 4x - 7$
4. $\lim_{x \rightarrow 0} 2x^3 - 3x^2 + 5x - 7$
5. $\lim_{x \rightarrow 1} x^3 - 4x^2 + 2x - 7$
6. $\lim_{x \rightarrow 2} 2x^3 + 3x - 7$

7. $\lim_{x \rightarrow 3} x^3 - x^2 + 4x - 7$

8. $\lim_{x \rightarrow 4} 2x^3 - 2x^2 + 5x - 7$

9. $\lim_{x \rightarrow -4} x^3 - 3x^2 + 2x - 7$

10. $\lim_{x \rightarrow 5} 2x^3 - 4x^2 + 3x - 7$

Level V: independent mastery

1. $\lim_{x \rightarrow -3} x^3 + 5x^2 + 14x + 7$

2. $\lim_{x \rightarrow -2} 2x^3 + 10x^2 + 28x + 15$

3. $\lim_{x \rightarrow -1} 3x^3 + 15x^2 + 42x + 23$

4. $\lim_{x \rightarrow 0} x^3 + 2x^2 + 20x + 7$

5. $\lim_{x \rightarrow 1} 2x^3 + 11x^2 + 26x + 19$

6. $\lim_{x \rightarrow 2} 3x^3 + 16x^2 + 40x + 27$

7. $\lim_{x \rightarrow 3} x^3 + 3x^2 + 18x + 11$

8. $\lim_{x \rightarrow 4} 2x^3 + 8x^2 + 32x + 19$

9. $\lim_{x \rightarrow -4} 3x^3 + 17x^2 + 38x + 31$

10. $\lim_{x \rightarrow 5} x^3 + 4x^2 + 16x + 15$

Topic 2

Rational Expressions by Direct Substitution

Learning target

Evaluate rational limits when the denominator is not zero at the target value.

Teaching notes

A rational expression is a fraction made from algebraic expressions. The main question is whether the denominator becomes zero at the target value. If the denominator does not become zero, the fraction is well-behaved there.

Big idea

Direct substitution works for rational expressions whenever the denominator is not 0. The denominator check comes first because division by zero is not allowed.

Procedure

1. Substitute the target value into the denominator.
2. If the denominator is not 0, substitute into the whole fraction.
3. Simplify the numerator and denominator separately before reducing.

Common trap

Do not cancel anything unless there is a reason to factor. When direct substitution works, just substitute and simplify.

Worked example

Problem. $\lim_{x \rightarrow -1} \frac{x^2 + 3x + 5}{x + 4}$

Move. Substitute because the denominator becomes 3, not 0.

Result. $\frac{1-3+5}{3} = 1$.

Drill

Level I: direct recognition

1. $\lim_{x \rightarrow -3} \frac{x-3}{x+10}$

2. $\lim_{x \rightarrow -2} \frac{2x-2}{x+11}$

3. $\lim_{x \rightarrow -1} \frac{3x-1}{x+12}$

4. $\lim_{x \rightarrow 0} \frac{4x}{x+13}$

5. $\lim_{x \rightarrow 1} \frac{x+1}{x+14}$

6. $\lim_{x \rightarrow 2} \frac{2x+2}{x+15}$

7. $\lim_{x \rightarrow 3} \frac{3x+3}{x+16}$

8. $\lim_{x \rightarrow 4} \frac{4x+4}{x+17}$

9. $\lim_{x \rightarrow -4} \frac{x+5}{x+18}$

10. $\lim_{x \rightarrow 5} \frac{2x+6}{x+19}$

Level II: routine variations

1. $\lim_{x \rightarrow -3} \frac{x^2 - 2x + 3}{x - 1}$

2. $\lim_{x \rightarrow -2} \frac{2x^2 - x + 3}{x - 3}$

3. $\lim_{x \rightarrow -1} \frac{3x^2 + 3}{x - 5}$

4. $\lim_{x \rightarrow 0} \frac{x^2 + x + 3}{x - 7}$

5. $\lim_{x \rightarrow 1} \frac{2x^2 + 2x + 3}{x - 9}$

6. $\lim_{x \rightarrow 2} \frac{3x^2 - 2x + 3}{x - 11}$

7. $\lim_{x \rightarrow 3} \frac{x^2 - x + 3}{x - 13}$

8. $\lim_{x \rightarrow 4} \frac{2x^2 + 3}{x - 15}$

9. $\lim_{x \rightarrow -4} \frac{3x^2 + x + 3}{x - 8}$

10. $\lim_{x \rightarrow 5} \frac{x^2 + 2x + 3}{x - 18}$

Level III: disguised or prepared forms

1. $\lim_{x \rightarrow -3} \frac{x^2 + 3x + 2}{x^2 + x - 56}$

2. $\lim_{x \rightarrow -2} \frac{x^2 + 3x + 2}{x^2 + x - 72}$

3. $\lim_{x \rightarrow -1} \frac{x^2 + 3x}{x^2 + x - 90}$

4. $\lim_{x \rightarrow 0} \frac{x^2 + 6x + 8}{x^2 + x - 110}$

5. $\lim_{x \rightarrow 1} \frac{x^2 + 2x + 1}{x^2 + 5x - 84}$

6. $\lim_{x \rightarrow 2} \frac{x^2 + 2x}{x^2 - 64}$

7. $\lim_{x \rightarrow 3} \frac{x^2 + 5x + 6}{x^2 - 81}$

8. $\lim_{x \rightarrow 4} \frac{x^2 + 5x + 4}{x^2 - 100}$

9. $\lim_{x \rightarrow -4} \frac{x^2 + x}{x^2 + 4x - 77}$

10. $\lim_{x \rightarrow 5} \frac{x^2 + 4x + 4}{x^2 + 4x - 96}$

Level IV: multi-stage execution

$$1. \lim_{x \rightarrow -3} \frac{\frac{x+1}{x+8}}{\frac{x-2}{x+11}}$$

$$2. \lim_{x \rightarrow -2} \frac{\frac{2x+1}{x+9}}{\frac{x-2}{x+12}}$$

$$3. \lim_{x \rightarrow -1} \frac{\frac{3x+1}{x+10}}{\frac{x-2}{x+13}}$$

$$4. \lim_{x \rightarrow 0} \frac{\frac{x+1}{x+11}}{\frac{x-2}{x+14}}$$

$$5. \lim_{x \rightarrow 1} \frac{\frac{2x+1}{x+12}}{\frac{x-2}{x+15}}$$

$$6. \lim_{x \rightarrow 2} \frac{\frac{3x+1}{x+13}}{\frac{x-2}{x+16}}$$

$$7. \lim_{x \rightarrow 3} \frac{\frac{x+1}{x+14}}{\frac{x-2}{x+11}}$$

$$8. \lim_{x \rightarrow 4} \frac{\frac{2x+1}{x+15}}{\frac{x-2}{x+12}}$$

$$9. \lim_{x \rightarrow -4} \frac{\frac{3x+1}{x+16}}{\frac{x-2}{x+13}}$$

$$10. \lim_{x \rightarrow 5} \frac{\frac{x+1}{x+17}}{\frac{x-2}{x+14}}$$

Level V: independent mastery

$$1. \lim_{x \rightarrow -3} \frac{x^3 - 2x - 5}{x^2 + 8x + 30}$$

$$2. \lim_{x \rightarrow -2} \frac{2x^3 - 2x - 4}{x^2 + 9x + 31}$$

$$3. \lim_{x \rightarrow -1} \frac{3x^3 - 2x - 3}{x^2 + 10x + 32}$$

$$4. \lim_{x \rightarrow 0} \frac{x^3 - 2x - 2}{x^2 + 11x + 33}$$

$$5. \lim_{x \rightarrow 1} \frac{2x^3 - 2x - 1}{x^2 + 8x + 34}$$

$$6. \lim_{x \rightarrow 2} \frac{3x^3 - 2x}{x^2 + 9x + 35}$$

$$7. \lim_{x \rightarrow 3} \frac{x^3 - 2x + 1}{x^2 + 10x + 36}$$

$$8. \lim_{x \rightarrow 4} \frac{2x^3 - 2x + 2}{x^2 + 11x + 37}$$

$$9. \lim_{x \rightarrow -4} \frac{3x^3 - 2x + 3}{x^2 + 8x + 38}$$

$$10. \lim_{x \rightarrow 5} \frac{x^3 - 2x + 4}{x^2 + 9x + 39}$$

Topic 3

Factoring and Cancellation Limits

Learning target

Handle $0/0$ rational limits by factoring completely, cancelling common factors, and then substituting.

Teaching notes

Sometimes direct substitution gives $\frac{0}{0}$. This does not mean the limit is 0, and it does not mean the limit is undefined. It means the expression needs more algebra before the limit can be evaluated.

Big idea

The form $\frac{0}{0}$ often signals a removable factor. Factor the numerator and denominator, cancel any common factor, and then evaluate the simpler expression. The cancellation does not say the original expression was defined at the target value; it says the nearby behavior can be represented by a simpler expression.

Procedure

1. Substitute first to see whether you get $\frac{0}{0}$.
2. Factor the numerator and denominator completely.
3. Cancel only entire common factors.
4. Substitute the target value into the simplified expression.

Common trap

Never cancel terms separated by addition or subtraction. For example, in $\frac{x+3}{x}$, the x in the numerator is not a factor by itself.

Worked example

Problem. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Move. Factor $x^2 - 9 = (x - 3)(x + 3)$, cancel the common factor, then substitute.

Result. 6.

Drill

Level I: direct recognition

1. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

2. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

3. $\lim_{x \rightarrow -5} \frac{x^2 - 25}{x + 5}$

4. $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$

5. $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4}$

6. $\lim_{x \rightarrow -3} \frac{2x^2 + 7x + 3}{x + 3}$

7. $\lim_{x \rightarrow 4} \frac{3x^2 - 11x - 4}{x - 4}$

8. $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$

9. $\lim_{x \rightarrow -3} \frac{x^3 + 27}{x + 3}$

10. $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2}$

Level II: routine variations

1. $\lim_{x \rightarrow 3} \frac{x^3 - 2x^2 - 9x + 18}{x^2 + 4x - 21}$

2. $\lim_{x \rightarrow 4} \frac{2x^3 - x^2 - 32x + 16}{x^2 + 4x - 32}$

3. $\lim_{x \rightarrow -5} \frac{3x^3 - 75x}{x^2 + 14x + 45}$

4. $\lim_{x \rightarrow -2} \frac{x^3 + 6x^2 + 11x + 6}{x^2 + 12x + 20}$

5. $\lim_{x \rightarrow 4} \frac{2x^3 - 26x - 24}{x^2 + 3x - 28}$

6. $\lim_{x \rightarrow -3} \frac{6x^3 + 27x^2 + 30x + 9}{x^2 + 11x + 24}$

7. $\lim_{x \rightarrow 4} \frac{3x^3 + x^2 - 48x - 16}{x^2 + 5x - 36}$

8. $\lim_{x \rightarrow 2} \frac{2x^4 + 5x^3 - 16x - 40}{x^2 + 8x - 20}$

9. $\lim_{x \rightarrow -3} \frac{3x^4 + 6x^3 + 81x + 162}{x^2 + 10x + 21}$

10. $\lim_{x \rightarrow 2} \frac{x^5 + 7x^4 - 16x - 112}{x^2 + 6x - 16}$

Level III: disguised or prepared forms

1. $\lim_{x \rightarrow 3} \frac{x^3 - 2x^2 - 6x + 9}{x^3 + 6x^2 - 9x - 54}$

2. $\lim_{x \rightarrow 4} \frac{2x^3 - 7x^2 - 7x + 12}{x^3 + 5x^2 - 16x - 80}$

3. $\lim_{x \rightarrow -5} \frac{3x^3 + 16x^2 + 2x - 15}{x^3 + 4x^2 - 25x - 100}$

4. $\lim_{x \rightarrow -2} \frac{4x^3 + 9x^2 - x - 6}{x^3 + 8x^2 + 21x + 18}$

5. $\lim_{x \rightarrow 4} \frac{x^3 - 3x^2 - 7x + 12}{x^3 + x^2 - 14x - 24}$

6. $\lim_{x \rightarrow -3} \frac{2x^3 + 7x^2 - 9}{2x^3 + 19x^2 + 45x + 18}$

7. $\lim_{x \rightarrow 4} \frac{3x^3 - 11x^2 - 7x + 12}{3x^3 + 4x^2 - 59x - 20}$

8. $\lim_{x \rightarrow 2} \frac{4x^3 - 7x^2 - 5x + 6}{x^4 + 4x^3 - 8x - 32}$

9. $\lim_{x \rightarrow -3} \frac{x^3 + 4x^2 - 9}{x^4 + 3x^3 + 27x + 81}$

10. $\lim_{x \rightarrow 2} \frac{2x^3 - 3x^2 - 5x + 6}{x^5 + 2x^4 - 16x - 32}$

Level IV: multi-stage execution

1. $\lim_{x \rightarrow 3} \frac{x^3 + x^2 - 9x - 9}{x^3 - x^2 - 5x - 3}$

2. $\lim_{x \rightarrow 4} \frac{x^3 + 2x^2 - 16x - 32}{x^3 - x^2 - 11x - 4}$

3. $\lim_{x \rightarrow -5} \frac{x^3 + 3x^2 - 25x - 75}{x^3 + 9x^2 + 21x + 5}$

4. $\lim_{x \rightarrow -2} \frac{x^3 + 9x^2 + 26x + 24}{x^3 + 7x^2 + 11x + 2}$

5. $\lim_{x \rightarrow 4} \frac{x^3 + 4x^2 - 17x - 60}{x^3 - 2x^2 - 7x - 4}$

6. $\lim_{x \rightarrow -3} \frac{2x^3 + 9x^2 + 10x + 3}{x^3 + 6x^2 + 10x + 3}$

7. $\lim_{x \rightarrow 4} \frac{3x^3 - 5x^2 - 26x - 8}{x^3 - 15x - 4}$

8. $\lim_{x \rightarrow 2} \frac{x^4 + 3x^3 - 8x - 24}{x^3 + 3x^2 - 9x - 2}$

9. $\lim_{x \rightarrow -3} \frac{x^4 + 4x^3 + 27x + 108}{x^3 + 5x^2 + 7x + 3}$

10. $\lim_{x \rightarrow 2} \frac{x^5 + 5x^4 - 16x - 80}{x^3 + x^2 - 5x - 2}$

Level V: independent mastery

1. $\lim_{x \rightarrow 3} \frac{x^3 - 2x^2 - 5x + 6}{x^3 + 5x^2 - 9x - 45}$

2. $\lim_{x \rightarrow 4} \frac{x^3 - 3x^2 - 10x + 24}{x^3 + 6x^2 - 16x - 96}$

3. $\lim_{x \rightarrow -5} \frac{x^3 + 6x^2 - 7x - 60}{x^3 + 7x^2 - 25x - 175}$

4. $\lim_{x \rightarrow -2} \frac{x^3 + 6x^2 + 3x - 10}{x^3 + 13x^2 + 46x + 48}$

5. $\lim_{x \rightarrow 4} \frac{x^3 - 4x^2 - 4x + 16}{x^3 + 8x^2 - 21x - 108}$

6. $\lim_{x \rightarrow -3} \frac{x^3 + 3x^2 - 9x - 27}{2x^3 + 27x^2 + 73x + 30}$

7. $\lim_{x \rightarrow 4} \frac{x^3 - x^2 - 16x + 16}{3x^3 + 4x^2 - 59x - 20}$

8. $\lim_{x \rightarrow 2} \frac{x^3 + x^2 - 16x + 20}{x^4 + 6x^3 - 8x - 48}$

9. $\lim_{x \rightarrow -3} \frac{x^3 + 2x^2 - 9x - 18}{x^4 + 7x^3 + 27x + 189}$

10. $\lim_{x \rightarrow 2} \frac{x^3 - 7x + 6}{x^5 + 8x^4 - 16x - 128}$

Topic 4

Left and Right Limits

Learning target

Evaluate left-hand and right-hand limits of piecewise functions by using the rule on the correct side of the target.

Teaching notes

A one-sided limit looks at the function from only one direction. The symbol $x \rightarrow a^-$ means x approaches a from the left, using values less than a . The symbol $x \rightarrow a^+$ means x approaches a from the right, using values greater than a .

Big idea

For a piecewise function, the rule you use depends on the side you are approaching from. A left-hand limit uses the formula attached to $x < a$. A right-hand limit uses the formula attached to $x > a$.

Procedure

1. Identify the target value.
2. For $x \rightarrow a^-$, use the piece of the function for inputs less than a .
3. For $x \rightarrow a^+$, use the piece of the function for inputs greater than a .
4. Substitute a into the correct side's rule.

Common trap

The value of $f(a)$, if it is even listed, does not control the one-sided limit. Limits care about nearby behavior.

Worked example

Problem. If $f(x) = \begin{cases} x + 1, & x < 2 \\ 3x, & x > 2 \end{cases}$, find both one-sided limits at $x = 2$.

Result. $\lim_{x \rightarrow 2^-} f(x) = 3$ and $\lim_{x \rightarrow 2^+} f(x) = 6$.

Drill

Level I: direct recognition

1. $f(x) = \begin{cases} x, & x < -3 \\ 2x, & x > -3 \end{cases}$; find $\lim_{x \rightarrow -3^-} f(x)$ and $\lim_{x \rightarrow -3^+} f(x)$
2. $f(x) = \begin{cases} x+1, & x < -2 \\ 2x-1, & x > -2 \end{cases}$; find $\lim_{x \rightarrow -2^-} f(x)$ and $\lim_{x \rightarrow -2^+} f(x)$
3. $f(x) = \begin{cases} x+2, & x < -1 \\ 2x-2, & x > -1 \end{cases}$; find $\lim_{x \rightarrow -1^-} f(x)$ and $\lim_{x \rightarrow -1^+} f(x)$
4. $f(x) = \begin{cases} x+3, & x < 0 \\ 2x-3, & x > 0 \end{cases}$; find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$
5. $f(x) = \begin{cases} x+4, & x < 1 \\ 2x-4, & x > 1 \end{cases}$; find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$
6. $f(x) = \begin{cases} x+5, & x < 2 \\ 2x-5, & x > 2 \end{cases}$; find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$
7. $f(x) = \begin{cases} x+6, & x < 3 \\ 2x-6, & x > 3 \end{cases}$; find $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$
8. $f(x) = \begin{cases} x+7, & x < 4 \\ 2x-7, & x > 4 \end{cases}$; find $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$
9. $f(x) = \begin{cases} x+8, & x < -4 \\ 2x-8, & x > -4 \end{cases}$; find $\lim_{x \rightarrow -4^-} f(x)$ and $\lim_{x \rightarrow -4^+} f(x)$
10. $f(x) = \begin{cases} x+9, & x < 5 \\ 2x-9, & x > 5 \end{cases}$; find $\lim_{x \rightarrow 5^-} f(x)$ and $\lim_{x \rightarrow 5^+} f(x)$

Level II: routine variations

1. $f(x) = \begin{cases} x^2, & x < -3 \\ -x, & x > -3 \end{cases}$; find both one-sided limits at $x = -3$
2. $f(x) = \begin{cases} x^2+1, & x < -2 \\ 3-x, & x > -2 \end{cases}$; find both one-sided limits at $x = -2$
3. $f(x) = \begin{cases} x^2+2, & x < -1 \\ 6-x, & x > -1 \end{cases}$; find both one-sided limits at $x = -1$
4. $f(x) = \begin{cases} x^2+3, & x < 0 \\ 9-x, & x > 0 \end{cases}$; find both one-sided limits at $x = 0$
5. $f(x) = \begin{cases} x^2+4, & x < 1 \\ 12-x, & x > 1 \end{cases}$; find both one-sided limits at $x = 1$
6. $f(x) = \begin{cases} x^2+5, & x < 2 \\ 15-x, & x > 2 \end{cases}$; find both one-sided limits at $x = 2$
7. $f(x) = \begin{cases} x^2+6, & x < 3 \\ 18-x, & x > 3 \end{cases}$; find both one-sided limits at $x = 3$
8. $f(x) = \begin{cases} x^2+7, & x < 4 \\ 21-x, & x > 4 \end{cases}$; find both one-sided limits at $x = 4$
9. $f(x) = \begin{cases} x^2+8, & x < -4 \\ 24-x, & x > -4 \end{cases}$; find both one-sided limits at $x = -4$
10. $f(x) = \begin{cases} x^2+9, & x < 5 \\ 27-x, & x > 5 \end{cases}$; find both one-sided limits at $x = 5$

Level III: disguised or prepared forms

1. $f(x) = \begin{cases} \frac{x^2-9}{x+3}, & x < -3 \\ x+2, & x > -3 \end{cases}$; find both one-sided limits at $x = -3$
2. $f(x) = \begin{cases} \frac{x^2-4}{x+2}, & x < -2 \\ 2x+2, & x > -2 \end{cases}$; find both one-sided limits at $x = -2$
3. $f(x) = \begin{cases} \frac{x^2-1}{x+1}, & x < -1 \\ 3x+2, & x > -1 \end{cases}$; find both one-sided limits at $x = -1$
4. $f(x) = \begin{cases} \frac{x^2+x}{x}, & x < 0 \\ x+2, & x > 0 \end{cases}$; find both one-sided limits at $x = 0$
5. $f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 2x+2, & x > 1 \end{cases}$; find both one-sided limits at $x = 1$
6. $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x < 2 \\ 3x+2, & x > 2 \end{cases}$; find both one-sided limits at $x = 2$
7. $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & x < 3 \\ x+2, & x > 3 \end{cases}$; find both one-sided limits at $x = 3$
8. $f(x) = \begin{cases} \frac{x^2-16}{x-4}, & x < 4 \\ 2x+2, & x > 4 \end{cases}$; find both one-sided limits at $x = 4$
9. $f(x) = \begin{cases} \frac{x^2-16}{x+4}, & x < -4 \\ 3x+2, & x > -4 \end{cases}$; find both one-sided limits at $x = -4$
10. $f(x) = \begin{cases} \frac{x^2-25}{x-5}, & x < 5 \\ x+2, & x > 5 \end{cases}$; find both one-sided limits at $x = 5$

Level IV: multi-stage execution

1. $f(x) = \begin{cases} x^2+6x+9, & x < -3 \\ \frac{x^2+2x+1}{x+9}, & x > -3 \end{cases}$; find $\lim_{x \rightarrow -3^-} f(x)$ and $\lim_{x \rightarrow -3^+} f(x)$
2. $f(x) = \begin{cases} x^2+4x+5, & x < -2 \\ \frac{x^2+x+1}{x+10}, & x > -2 \end{cases}$; find $\lim_{x \rightarrow -2^-} f(x)$ and $\lim_{x \rightarrow -2^+} f(x)$
3. $f(x) = \begin{cases} x^2+2x+3, & x < -1 \\ \frac{x^2+1}{x+11}, & x > -1 \end{cases}$; find $\lim_{x \rightarrow -1^-} f(x)$ and $\lim_{x \rightarrow -1^+} f(x)$
4. $f(x) = \begin{cases} x^2+3, & x < 0 \\ \frac{x^2-x+1}{x+12}, & x > 0 \end{cases}$; find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$
5. $f(x) = \begin{cases} x^2-2x+5, & x < 1 \\ \frac{x^2+2x+1}{x+13}, & x > 1 \end{cases}$; find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$
6. $f(x) = \begin{cases} x^2-4x+9, & x < 2 \\ \frac{x^2+x+1}{x+14}, & x > 2 \end{cases}$; find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$
7. $f(x) = \begin{cases} x^2-6x+15, & x < 3 \\ \frac{x^2+1}{x+15}, & x > 3 \end{cases}$; find $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$
8. $f(x) = \begin{cases} x^2-8x+23, & x < 4 \\ \frac{x^2-x+1}{x+16}, & x > 4 \end{cases}$; find $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$
9. $f(x) = \begin{cases} x^2+8x+24, & x < -4 \\ \frac{x^2+2x+1}{x+17}, & x > -4 \end{cases}$; find $\lim_{x \rightarrow -4^-} f(x)$ and $\lim_{x \rightarrow -4^+} f(x)$
10. $f(x) = \begin{cases} x^2-10x+34, & x < 5 \\ \frac{x^2+x+1}{x+18}, & x > 5 \end{cases}$; find $\lim_{x \rightarrow 5^-} f(x)$ and $\lim_{x \rightarrow 5^+} f(x)$

Level V: independent mastery

1. $f(x) = \begin{cases} \frac{x^2+4x+3}{x+3}, & x < -3 \\ \frac{2x^2+6x}{x+3}, & x > -3 \end{cases}$; find both one-sided limits at $x = -3$
2. $f(x) = \begin{cases} \frac{x^2+4x+4}{x+2}, & x < -2 \\ \frac{2x^2+3x-2}{x+2}, & x > -2 \end{cases}$; find both one-sided limits at $x = -2$

3. $f(x) = \begin{cases} \frac{x^2+4x+3}{x+1}, & x < -1 \\ \frac{2x^2-2}{x+1}, & x > -1 \end{cases}$; find both one-sided limits at $x = -1$
4. $f(x) = \begin{cases} \frac{x^2+4x}{x}, & x < 0 \\ \frac{2x^2-3x}{x}, & x > 0 \end{cases}$; find both one-sided limits at $x = 0$
5. $f(x) = \begin{cases} \frac{x^2+4x-5}{x-1}, & x < 1 \\ \frac{2x^2-6x+4}{x-1}, & x > 1 \end{cases}$; find both one-sided limits at $x = 1$
6. $f(x) = \begin{cases} \frac{x^2+4x-12}{x-2}, & x < 2 \\ \frac{2x^2-9x+10}{x-2}, & x > 2 \end{cases}$; find both one-sided limits at $x = 2$
7. $f(x) = \begin{cases} \frac{x^2+4x-21}{x-3}, & x < 3 \\ \frac{2x^2-12x+18}{x-3}, & x > 3 \end{cases}$; find both one-sided limits at $x = 3$
8. $f(x) = \begin{cases} \frac{x^2+4x-32}{x-4}, & x < 4 \\ \frac{2x^2-15x+28}{x-4}, & x > 4 \end{cases}$; find both one-sided limits at $x = 4$
9. $f(x) = \begin{cases} \frac{x^2+13x+36}{x+4}, & x < -4 \\ \frac{2x^2-32}{x+4}, & x > -4 \end{cases}$; find both one-sided limits at $x = -4$
10. $f(x) = \begin{cases} \frac{x^2+5x-50}{x-5}, & x < 5 \\ \frac{2x^2-19x+45}{x-5}, & x > 5 \end{cases}$; find both one-sided limits at $x = 5$

Topic 5

One-Sided Limits That Disagree

Learning target

Compare one-sided limits and conclude that the two-sided limit does not exist when they disagree.

Teaching notes

A two-sided limit exists only when the left-hand and right-hand limits approach the same value. If the two sides approach different values, the function has a jump at that point, and the two-sided limit does not exist.

Big idea

The two sides must agree. It is not enough for each one-sided limit to exist separately. They have to land on the same number.

Procedure

1. Find $\lim_{x \rightarrow a^-} f(x)$ using the left-hand rule.
2. Find $\lim_{x \rightarrow a^+} f(x)$ using the right-hand rule.
3. Compare the two answers.
4. If they are equal, the two-sided limit is that shared value. If they are different, write that the limit does not exist.

Common trap

Do not average the two one-sided limits or choose the side that looks simpler. Disagreement means the two-sided limit does not exist.

Worked example

Problem. If $f(x) = \begin{cases} x, & x < 1 \\ x + 4, & x > 1 \end{cases}$, decide whether $\lim_{x \rightarrow 1} f(x)$ exists.

Result. The left-hand limit is 1 and the right-hand limit is 5, so the two-sided limit does not exist.

Drill

Level I: direct recognition

1. $f(x) = \begin{cases} x, & x < -3 \\ x+3, & x > -3 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -3} f(x)$ exists
2. $f(x) = \begin{cases} x+1, & x < -2 \\ x+4, & x > -2 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -2} f(x)$ exists
3. $f(x) = \begin{cases} x+2, & x < -1 \\ x+5, & x > -1 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -1} f(x)$ exists
4. $f(x) = \begin{cases} x+3, & x < 0 \\ x+6, & x > 0 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 0} f(x)$ exists
5. $f(x) = \begin{cases} x+4, & x < 1 \\ x+7, & x > 1 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 1} f(x)$ exists
6. $f(x) = \begin{cases} x+5, & x < 2 \\ x+8, & x > 2 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 2} f(x)$ exists
7. $f(x) = \begin{cases} x+6, & x < 3 \\ x+9, & x > 3 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 3} f(x)$ exists
8. $f(x) = \begin{cases} x+7, & x < 4 \\ x+10, & x > 4 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 4} f(x)$ exists
9. $f(x) = \begin{cases} x+8, & x < -4 \\ x+11, & x > -4 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -4} f(x)$ exists
10. $f(x) = \begin{cases} x+9, & x < 5 \\ x+12, & x > 5 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 5} f(x)$ exists

Level II: routine variations

1. $f(x) = \begin{cases} 2x, & x < -3 \\ -x, & x > -3 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -3$
2. $f(x) = \begin{cases} 2x-1, & x < -2 \\ 1-x, & x > -2 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -2$
3. $f(x) = \begin{cases} 2x-2, & x < -1 \\ 2-x, & x > -1 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -1$
4. $f(x) = \begin{cases} 2x-3, & x < 0 \\ 3-x, & x > 0 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 0$
5. $f(x) = \begin{cases} 2x-4, & x < 1 \\ 4-x, & x > 1 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 1$
6. $f(x) = \begin{cases} 2x-5, & x < 2 \\ 5-x, & x > 2 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 2$
7. $f(x) = \begin{cases} 2x-6, & x < 3 \\ 6-x, & x > 3 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 3$
8. $f(x) = \begin{cases} 2x-7, & x < 4 \\ 7-x, & x > 4 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 4$
9. $f(x) = \begin{cases} 2x-8, & x < -4 \\ 8-x, & x > -4 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -4$
10. $f(x) = \begin{cases} 2x-9, & x < 5 \\ 9-x, & x > 5 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 5$

Level III: disguised or prepared forms

1. $f(x) = \begin{cases} \frac{x^2+5x+6}{x+3}, & x < -3 \\ x+5, & x > -3 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -3} f(x)$ exists
2. $f(x) = \begin{cases} \frac{x^2+5x+6}{x+2}, & x < -2 \\ x+6, & x > -2 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -2} f(x)$ exists
3. $f(x) = \begin{cases} \frac{x^2+5x+4}{x+1}, & x < -1 \\ x+7, & x > -1 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -1} f(x)$ exists
4. $f(x) = \begin{cases} \frac{x^2+5x}{x}, & x < 0 \\ x+8, & x > 0 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 0} f(x)$ exists
5. $f(x) = \begin{cases} \frac{x^2+5x-6}{x-1}, & x < 1 \\ x+9, & x > 1 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 1} f(x)$ exists
6. $f(x) = \begin{cases} \frac{x^2+5x-14}{x-2}, & x < 2 \\ x+10, & x > 2 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 2} f(x)$ exists
7. $f(x) = \begin{cases} \frac{x^2+5x-24}{x-3}, & x < 3 \\ x+11, & x > 3 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 3} f(x)$ exists
8. $f(x) = \begin{cases} \frac{x^2+5x-36}{x-4}, & x < 4 \\ x+12, & x > 4 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 4} f(x)$ exists
9. $f(x) = \begin{cases} \frac{x^2+14x+40}{x+4}, & x < -4 \\ x+13, & x > -4 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -4} f(x)$ exists
10. $f(x) = \begin{cases} \frac{x^2+6x-55}{x-5}, & x < 5 \\ x+14, & x > 5 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 5} f(x)$ exists

Level IV: multi-stage execution

1. $f(x) = \begin{cases} x^2+6x+9, & x < -3 \\ x^2+6x+13, & x > -3 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -3$
2. $f(x) = \begin{cases} x^2+4x+5, & x < -2 \\ x^2+4x+9, & x > -2 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -2$
3. $f(x) = \begin{cases} x^2+2x+3, & x < -1 \\ x^2+2x+7, & x > -1 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -1$
4. $f(x) = \begin{cases} x^2+3, & x < 0 \\ x^2+7, & x > 0 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 0$
5. $f(x) = \begin{cases} x^2-2x+5, & x < 1 \\ x^2-2x+9, & x > 1 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 1$
6. $f(x) = \begin{cases} x^2-4x+9, & x < 2 \\ x^2-4x+13, & x > 2 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 2$
7. $f(x) = \begin{cases} x^2-6x+15, & x < 3 \\ x^2-6x+19, & x > 3 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 3$
8. $f(x) = \begin{cases} x^2-8x+23, & x < 4 \\ x^2-8x+27, & x > 4 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 4$
9. $f(x) = \begin{cases} x^2+8x+24, & x < -4 \\ x^2+8x+28, & x > -4 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -4$
10. $f(x) = \begin{cases} x^2-10x+34, & x < 5 \\ x^2-10x+38, & x > 5 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 5$

Level V: independent mastery

1. $f(x) = \begin{cases} \frac{x^2-9}{x+3}, & x < -3 \\ 2x+1, & x > -3 \end{cases}$; find both one-sided limits and decide whether the limit exists
2. $f(x) = \begin{cases} \frac{x^2-4}{x+2}, & x < -2 \\ 2x+2, & x > -2 \end{cases}$; find both one-sided limits and decide whether the limit exists

3. $f(x) = \begin{cases} \frac{x^2-1}{x+1}, & x < -1 \\ 2x+3, & x > -1 \end{cases}$; find both one-sided limits and decide whether the limit exists
4. $f(x) = \begin{cases} \frac{x^2+x}{x}, & x < 0 \\ 2x+4, & x > 0 \end{cases}$; find both one-sided limits and decide whether the limit exists
5. $f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 2x+5, & x > 1 \end{cases}$; find both one-sided limits and decide whether the limit exists
6. $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x < 2 \\ 2x+6, & x > 2 \end{cases}$; find both one-sided limits and decide whether the limit exists
7. $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & x < 3 \\ 2x+7, & x > 3 \end{cases}$; find both one-sided limits and decide whether the limit exists
8. $f(x) = \begin{cases} \frac{x^2-16}{x-4}, & x < 4 \\ 2x+8, & x > 4 \end{cases}$; find both one-sided limits and decide whether the limit exists
9. $f(x) = \begin{cases} \frac{x^2-16}{x+4}, & x < -4 \\ 2x+9, & x > -4 \end{cases}$; find both one-sided limits and decide whether the limit exists
10. $f(x) = \begin{cases} \frac{x^2-25}{x-5}, & x < 5 \\ 2x+10, & x > 5 \end{cases}$; find both one-sided limits and decide whether the limit exists

Topic 6

Limit Algebra Laws and Reciprocals

Learning target

Use sum, difference, product, quotient, power, root, and reciprocal limit laws from known component limits.

Teaching notes

Limit laws let you build a complicated limit from simpler known limits. If you know what $f(x)$, $g(x)$, and $h(x)$ approach, you can usually combine those approaching values using ordinary arithmetic.

Big idea

Limits respect addition, subtraction, multiplication, powers, and many roots. Quotients and reciprocals also work, but only when the denominator's limit is not 0.

Procedure

1. Replace each function with its given limit value.
2. Keep the same operations from the original expression.
3. Simplify the resulting number expression.
4. Check that no denominator limit is 0 before using a quotient or reciprocal law.

Common trap

The reciprocal law says $\lim_{x \rightarrow c} \frac{1}{f(x)} = \frac{1}{\lim_{x \rightarrow c} f(x)}$ only when $\lim_{x \rightarrow c} f(x) \neq 0$.

Worked example

Problem. Suppose $\lim_{x \rightarrow c} f(x) = 2$ and $\lim_{x \rightarrow c} g(x) = -3$. Find $\lim_{x \rightarrow c} (f(x)g(x) + 4)$.

Result. $2(-3) + 4 = -2$.

Drill

Level I: direct recognition

1. Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} f(x) + g(x)$.
2. Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} f(x) - g(x)$.
3. Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} 3f(x) + 2g(x)$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} f(x)g(x)$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{1}{f(x)}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} f(x)^2 + g(x)$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} h(x) - f(x)g(x)$.
9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} \frac{f(x) + h(x)}{g(x)}$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{g(x)h(x)}{f(x)}$.

Level II: routine variations

1. Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} 2f(x) - 3g(x) + h(x)$.
2. Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{h(x)}{g(x) - f(x)}$.
3. Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} (f(x) + g(x))^2$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} \frac{1}{g(x)^2}$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} f(x)^2 - g(x)^2$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{f(x)g(x) + h(x)}{2}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{h(x) - g(x)}{f(x) + 1}$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} \frac{f(x) + 2}{h(x) - 1}$.
9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} \frac{f(x)^2 + h(x)}{g(x)}$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{1}{f(x)g(x)}$.

Level III: disguised or prepared forms

1. Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} \sqrt{h(x) + 5}$.
2. Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \sqrt{f(x)^2 + g(x)^2}$.
3. Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} \frac{\sqrt{h(x) + 12}}{g(x)}$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} f(x)^3 - 2h(x)$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} \frac{4}{h(x)}$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{f(x) - g(x)}{h(x) + 2}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{f(x)g(x)}{h(x)}$.

- $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{h(x)^2}{f(x) - g(x)}$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} \frac{f(x) + g(x) + h(x)}{3}$.
9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} (2f(x) - g(x))h(x)$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{1}{h(x) - f(x)}$.

Level IV: multi-stage execution

1. Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} \frac{f(x)^2 + g(x)^2}{h(x)}$.
2. Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \sqrt{h(x)^2 + f(x)^2}$.
3. Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} \frac{g(x)}{f(x)^2 + 1}$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} \frac{h(x) - 2g(x)}{f(x) + 3}$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} \frac{f(x)h(x)}{g(x)^2}$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{1}{\sqrt{h(x) + 9}}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \sqrt{f(x) + g(x) + 10}$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} \frac{f(x)^2 - g(x)}{h(x) - g(x)}$.
9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} \left(\frac{f(x)}{g(x)} \right)^2$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{h(x) + 1}{f(x)g(x) - 2}$.

Level V: independent mastery

1. Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} \frac{2}{f(x)} + \frac{3}{g(x)}$.
2. Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} + \frac{g(x)}{h(x)}$.
3. Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} \frac{h(x)}{f(x)} - \frac{f(x)}{h(x)}$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} \frac{f(x)^2}{g(x)} + h(x)$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} \frac{1}{f(x) + g(x)}$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{f(x) + g(x)}{f(x) - g(x)}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \sqrt{h(x) + f(x)^2}$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} \frac{h(x)^2 - f(x)^2}{g(x)}$.
9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} \frac{f(x) + 1}{g(x) - 1}$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{h(x) + g(x)}{f(x)^2 + 2}$.

Before Declaring the Limit Complete

- I checked whether direct substitution is allowed.
- I factored before cancelling in any $0/0$ rational limit.
- I cancelled only common factors, never separate terms.
- I used the left-hand rule for $x \rightarrow a^-$ and the right-hand rule for $x \rightarrow a^+$.
- I concluded DNE only after comparing the two one-sided limits.
- I avoided the quotient and reciprocal laws when the denominator limit is 0.